Heat Transfer during Nd:Yag Pulsed Laser Welding and Its Effect on Solidification Structure of Austenitic Stainless Steels

T. ZACHARIA, S.A. DAVID, J.M. VITEK, and T. DEBROY

Theoretical and experimental investigations were carried out to determine the effect of process parameters on weld metal microstructures of austenitic stainless steels during pulsed laser welding. Laser welds made on four austenitic stainless steels at different power levels and scanning speeds were considered. A transient heat transfer model that takes into account fluid flow in the weld pool was employed to simulate thermal cycles and cooling rates experienced by the material under various welding conditions. The weld metal thermal cycles and cooling rates are related to features of the solidification structure. For the conditions investigated, the observed fusion zone structure ranged from duplex austenite (γ) + ferrite (δ) to fully austenitic or fully ferritic. Unlike welding with a continuous wave laser, pulsed laser welding results in thermal cycling from multiple melting and solidification cycles in the fusion zone, causing significant post-solidification solid-state transformation to occur. There was microstructural evidence of significant recrystallization in the fusion zone structure that can be explained on the basis of the thermal cycles. The present investigation clearly demonstrated the potential of the computational model to provide detailed information regarding the heat transfer conditions experienced during welding.

I. INTRODUCTION

ONE of the major problems encountered during fusion joining of austenitic stainless steels is the propensity for hot-cracking.[1,2] One way of controlling the crack sensitivity of austenitic stainless steel welds is by adjusting the composition so as to produce approximately 5 pct delta (δ) ferrite in the weld metals.[3] It is well recognized that the amount of ferrite is related to the composition of the weld metal. To predict the amount of delta ferrite in the weld metal microstructure, Schaeffler[4] developed a constitution diagram in terms of the important austenite stabilizers and ferrite stabilizers in the weld deposit. This diagram was later modified by Delong[5] to introduce the austenite stabilizing effect of nitrogen. However, the ferrite morphology (size, shape, and distribution) in the weld metal microstructure often exhibits considerable variation and is not addressed in these constitution diagrams.

The rate at which the weldment cools significantly influences the ferrite morphology and distribution.[6] It is well documented that, for GTA welds, the ferrite morphology is strongly dependent on the process parameters.[7,8] For high heat inputs with relatively low cooling rates, the solidification substructure tends to be coarse and results in a widely spaced ferrite network. Lower heat inputs, on the other hand, produce a finer ferrite network. In the case of high energy beam processes such as laser welding, other types of microstructural modification have been found. For example, changes in the mode of solidification were observed after laser welding.[9-12] As a result, the rapid solidification of the weld metal during laser welding makes predictions of ferrite morphology from conventional constitution diagrams impossible.

Recognizing the need for including the effect of cooling rate, David *et al.*,[12] in a recent publication, proposed modifications to the Schaeffler constitution diagram to incorporate the effect of rapid solidification on weld metal microstructure. The authors showed that, during laser welding, both weld pool cooling rates and post-solidification solid-state cooling rates can significantly influence the weld metal microstructure by changing the primary mode of solidification and/or suppressing subsequent solid-state transformation. Since cooling rates are sensitive to power density and scanning speed of the laser, it is of considerable interest to describe quantitatively the effect of process parameters on the solidification behavior and the resultant structure.

Accurate measurement of puddle temperature and cooling rates during laser welding are extremely difficult due to the small size of the weld pool and large temperature gradients and cooling rates involved. In the previous report,[12] the authors estimated the values of cooling rates on the basis of the solution of a relatively simple, two-dimensional heat conduction equation.[13] While such analytical equations represent a simple and convenient method of determining cooling rates at specified temperatures, they involve several simplifications that cannot be justified on the basis of the physics of the process. In particular, the model was based on the assumption of a point heat source and, more importantly, the effects of latent heat of fusion and convective heat transfer were ignored. Perhaps the biggest disadvantage in using such analytical techniques, particularly in light of the present analysis, is that the model cannot consider the effect of pulsed laser welding. Intermittent laser beam–metal interaction can cause extremely complex thermal cycles,

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making prediction of cooling rates using such analytical techniques virtually impossible. An alternative method for obtaining this information is by numerically solving mathematical models that represent the physical features of the process.

In recent years, our understanding of the laser welding process and our ability to model the physical phenomena has improved tremendously. Several mathematical models[14–18] have been developed to provide insight into the complex phenomena that occur during welding processes. However, reliable experimental values of several key parameters, such as reflectivity and thermophysical properties, are not available yet. Nonetheless, it is felt that computational modeling can be a useful tool in obtaining invaluable information that can further our understanding of process-property relationships.

In this paper, a computational model for pulsed laser welding is applied to understand the solidification structure after pulsed laser welding of austenitic stainless steels. Numerically computed temperatures and cooling rates at critical temperatures are correlated with the experimentally observed solidification structure.

II. EXPERIMENTAL PROCEDURE

The materials used in this investigation and their compositions are given in Table I. These alloys were arc melted, drop cast into rectangular ingots, and cold rolled to 0.51 mm thick sheets. Laser beam welds were made on 50 × 50 mm coupons in an argon atmosphere (14 liters/minute) using a Raytheon, Model SS-500, pulsed Nd:YAG laser. Welding was performed with the laser beam focused at the metal surface using a 15.24 cm focal length lens with an antireflection coating. A pulse length of 1 ms and a pulse rate of 200 pps were used. In order to determine the effect of heat input and scanning speed on the cooling rate and, hence, the solidification structure, two power levels of 200 and 120 watts and five scanning speeds ranging from 12.7 to 254 cm/minute (5 to 100 ipm) were used.

Microstructural characterization was done by optical and transmission electron microscopy. Longitudinal and transverse sections of the weld metal from all the heats were prepared for optical metallography. Specimens were prepared using standard procedures and then etched using a solution containing HNO_3 and H_2O.

Table I. Alloy Composition (Wt Pct)

Element	304	308	310	312	316
Cr	18.16	19.17	25.62	28.92	19.25
Ni	8.63	10.57	19.18	8.44	12.73
Mn	1.28	1.92	0.90	1.24	2.22
C	0.06	0.06	0.03	0.13	0.03
Si	0.72	0.72	0.42	0.43	0.45
P	0.032	0.045	0.013	0.018	0.042
S	0.007	0.016	0.005	0.004	0.018
Mo	—	—	0.15	0.15	2.34
Ti	—	0.56	—	—	0.30
Cr_{eq}	19.24	20.25	26.40	29.72	22.27
Ni_{eq}	11.07	13.33	20.53	12.96	14.74
Cr_{eq}/Ni_{eq}	1.74	1.52	1.29	2.29	1.51

III. THEORETICAL INVESTIGATION

During pulsed laser welding, the base metal is irradiated intermittently with a high energy laser beam causing rapid melting and solidification within a very small area. The weld bead is made by a technique of overlapping spot welds. Hence, every location in the weld metal experiences successive thermal cycles. Each thermal cycle differs from the previously experienced thermal cycle, since the distance from the beam-metal interaction continuously changes as the laser beam traverses along the direction of the weld. Another factor that alters the thermal cycle with each laser-metal interaction is the change in temperature of the material and corresponding temperature-dependent properties of the material at the beginning of the pulse. It is important to understand the detailed phenomena that take place during and after each laser metal interaction in order to study its effect on the solidification structure. A mathematical model that describes the pulsed laser welding process was solved numerically to obtain information regarding the thermal cycles and cooling rates that occur during the welding conditions.

A. *Model for Pulsed Laser Welding*

Figure 1 is a schematic drawing of the laser welding process showing the region of interest. The metal surface is irradiated with a laser beam and a molten puddle is formed due to the energy transfer between the beam and the metal surface. For laser welding, the primary driving force for convection in the molten pool is the force due to interfacial tension gradients prevailing on the surface. The importance of convective heat transfer in determining the fusion zone geometry and local temperature field has been demonstrated by several theoretical[14–18] and experimental investigations.[19] Therefore, any model used to predict the temperature cycle and cooling rates that occur during pulsed laser welding must incorporate convective heat transfer.

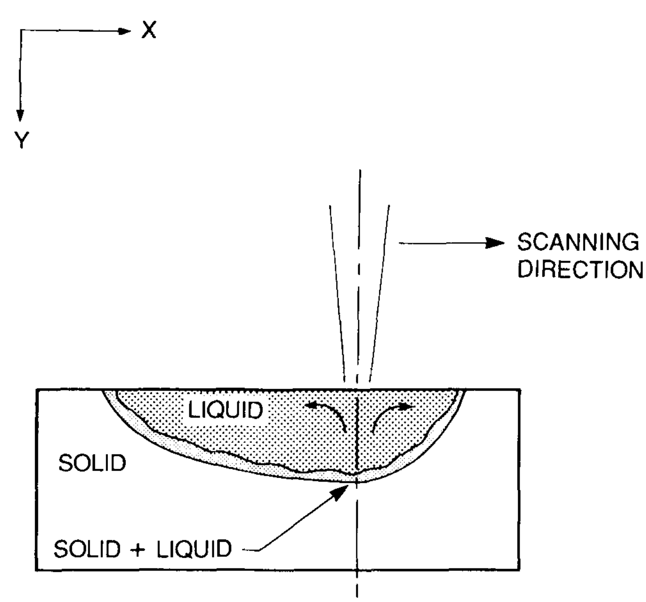

Fig. 1 — Schematic drawing of the laser welding process, indicating the region of interest.

A computational model,[17] using the finite difference analysis technique (FDA), was modified appropriately to investigate the heat transfer conditions during Nd:YAG pulsed laser welding of austenitic stainless steels. The model solves steady and transient two-dimensional equations that describe the flow and heat transfer conditions in the fusion zone.

The following assumptions were made in the formulation of the model:

(1) The fluid flow and heat transfer inside the molten pool are adequately described by a two-dimensional, flat surface, constant density representation, along the scanning direction (x-axis) of the laser (Figure 1).
(2) The flow is driven primarily by the surface tension gradient. The surface tension gradient was considered a constant.
(3) The power distribution of the heat source is considered as Gaussian, based on available literature.[20,21,22]
(4) The effect of vaporization on convection is ignored.

B. *Mathematical Formulation*

The definitions of all the symbols are presented in the appendix.

1. *Conservation of mass*

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \qquad [1]$$

2. *Conservation of momentum*

a. *X-direction*

$$\frac{\partial u}{\partial t} + \frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + g_x + \nu\left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right] \qquad [2a]$$

b. *Y-direction*

$$\frac{\partial v}{\partial t} + \frac{\partial uv}{\partial x} + \frac{\partial v^2}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial y} + g_y + \nu\left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right] \qquad [2b]$$

3. *Conservation of energy*

The enthalpy, rather than the temperature, was considered as the dependent variable in the energy equation. Here, the latent heat is included in the definition of enthalpy. The energy equation in two dimensions may be expressed as:

$$\frac{\partial H}{\partial t} + u\frac{\partial H}{\partial x} + v\frac{\partial H}{\partial y} = \frac{\kappa}{\rho C_p}\left[\frac{\partial^2 H}{\partial x^2} + \frac{\partial^2 H}{\partial y^2}\right] + S_H(x, t) \qquad [3a]$$

The source term in the above equation, $S_H(x, t)$, represents the absorption of heat from the laser beam. The heat flux from the laser beam, $\mathbf{q}_s(x, t)$, was described by a Gaussian distribution and was given by the following equation:

$$\mathbf{q}_s(x, t) = \delta_k \frac{3Q}{\pi r_b^2} e^{-[3x^2/r_b^2]} \qquad [3b]$$

$$\delta_k = 1 \quad \text{for pulse on} \qquad [3c]$$

$$\delta_k = 0 \quad \text{for pulse off} \qquad [3d]$$

The local value of the source term was related to the heat flux through the following equation:

$$S_H(x, t) = \frac{\varepsilon}{\rho y_d} \mathbf{q}_s(x, t) \qquad [3e]$$

The fraction of incident energy absorbed is represented by the absorptivity, ε, of the material.

The above equation is used to calculate the temperature distribution of the workpiece, with the help of a relationship for enthalpy as a function of temperature covering both the solid and liquid phases.

C. *Thermal Boundary Conditions*

(1) At the surface of the sample the source term is zero except for the region which is directly under the heat source.
(2) The heat transfer between the surface of the sample and the surroundings is a specified input parameter.
(3) The heat transfer at the bottom and the sides of the plate is calculated by equating the conduction heat flux with the heat flux due to convection from the plate surface.

D. *Dynamic Boundary Conditions*

(1) Along the solid-liquid interface, the liquid velocity is zero as no-slip conditions are imposed.
(2) At the surface of the pool, the surface tension variation with temperature must be balanced by fluid shear since the surface must be continuous. Therefore, the shear stress at the surface is equated to the gradient of surface tension.

An outline of the solution procedure, information on the grids used, a discussion of stability, convergence, and the accuracy of the solution are presented in the appendix. The data used for the computations are given in Table II.

IV. RESULTS AND DISCUSSION

In the pulse mode, the laser metal interaction takes place intermittently for short durations (in this case, 1 ms). Due to the nature of the process, the weld metal experiences several cycles of rapid melting and solidification before the final solidification. The peak temperature and cooling rate at each location, parameters that control the solidification structure, are largely de-

Table II. Data Used for Heat Transfer Calculations

Laser power	120, 200 W
Absorptivity[24]	30 pct
Effective radius of heat source	0.15 mm
Liquidus temperature	1725 K
Heat of fusion	0.634×10^2 cal/gm
Boiling temperature	3075 K
Density	7.6 gm/cm^3
Temperature coefficient of surface tension[27]	-0.1 gm/sec^2 K
Effective viscosity of molten metal	5 gm/cm sec
Effective thermal conductivity of molten metal	0.37 cal/cm sec K
Heat transfer coefficient at the solid surface	$3.15 \times 10^{-4}\,\Delta T^{0.25}$ cal/sec cm^2 K

pendent on (1) the absorptivity of the material; (2) the energy (power) density; (3) the pulse duration of the laser; (4) the scanning speed of the laser beam. The absorptivity of the metal surface is a material property that depends on a number of variables such as the nature of the surface, the level of oxidation, the temperature, and the wavelength and power density of incident radiation.[23] For the present analysis, a laser absorptivity value of 30 pct was used, based on data reported in the literature[24] for Nd:YAG lasers.

A. *Weld Thermal Cycle*

The detailed information on the peak temperatures and the temperature variation that a material experiences during laser welding is of considerable metallurgical importance. In attempting to calculate the thermal cycle during laser welding, the approach used must describe the actual process accurately. The temperature reached at any location is a function of the heat input, the thermophysical properties of the material, and the prior temperature of the material. The temperature at any location in a specimen would be increased considerably if the base metal were to be preheated prior to welding. During pulsed laser welding, the base metal is preheated by the previous laser metal interactions. Hence, in order to accurately determine the temporal and spatial variation of the temperature and its effect on the metallurgical properties of the weldment, the pulsing effect of the laser was incorporated in the calculations.

The temperature experienced by a point on the surface of the metal, 8.35 mm away from the edge of the specimen along the scanning line (x-axis), was calculated as a function of time for all the welding conditions. Since the heat source is not continuous (pulsed laser), the precise nature of the calculated thermal cycle and the magnitude of temperature depend on the exact relation between the monitoring point and the location of the laser-metal interaction. For example, at the welding speed of 254 cm/minute, the laser travels 0.02 cm between pulses. At this speed, it is possible for the laser beam to interact with the material at distances of 0.01 cm on either side of the monitoring point. Since the location of each laser-metal interaction is different for different welding speeds, to be consistent, the starting point was adjusted for each welding speed so that at least one laser-metal interaction took place directly at the monitoring point.

Figures 2 and 3 show theoretically predicted temperatures at the monitoring point for power levels of 200 and 120 watts, respectively, as a function of laser speed. The results indicate that, at any location on the metal surface, the material is subjected to several temperature cycles during welding in the pulsed mode. The number of melting cycles initially increases with increasing power and decreasing speed. At the slower speeds, the overlap of the thermal effects of successive laser pulses increases considerably, causing the overall temperature of the weld metal to increase so that after sufficient time has elapsed, the weld metal remains molten even during the off time of the laser beam. Therefore, for the slowest welding speeds, the intermittent heat source produced a thermal history similar to that expected from a continuous wave laser. For the high heat input, this effect was observed for the two slowest speeds (50.4, 12.7 cm/minute), as shown in Figures 2(d) and (e). For the low heat input, however, this effect was observed only for the slowest speed (Figure 3(e)).

The predicted thermal cycles and temperatures may vary with the location of the monitoring point since the heat source is not continuous. In order to determine the influence of the monitoring location on the observed thermal history, the monitoring point was shifted by 0.06 mm along the scanning direction from the previous monitoring location. The thermal history at this new monitoring location was calculated for the same welding conditions (200 watts, 190 cm/minute), as in Figure 2(b) and is shown in Figure 4. Comparison of Figure 4 and Figure 2(b) shows noticeable differences in the thermal behavior of the weld metal at the two locations. At this new location, the results indicate that the magnitude of the first two temperature peaks decreased, whereas the magnitude of the third temperature peak increased. In addition, a fourth post-solidification temperature peak was observed. This result indicates that the thermal cycle experienced at every location is unique for each welding condition. Nonetheless, the difference in the thermal behavior for the different locations was less than the difference found at the different welding speeds.

B. *Calculated Cooling Rates*

The final solidification structure and mechanical properties of the weld zone are determined primarily by the cooling rates at several critical temperatures corresponding to the liquid-solid transformation and the solid-solid transformations. Most alloys solidify with a dendritic structure during welding. The grain size and the spacings between the dendrite arms are determined by the cooling rates. Rapid cooling rates or shorter solidification times result in smaller dendrite arm spacings and relatively finer structures. Hence, the importance of obtaining reliable and complete information about the cooling rates throughout the temperature range of interest cannot be overemphasized. One of the major hurdles in developing a relationship between processing conditions and properties is the difficulty associated with experimental measurement of the cooling rates. Therefore, computational modeling of the process provides a very valuable tool in calculating the cooling rates and the temporal and spatial variation of cooling rates.

Cooling rates for all simulations considered in this paper were calculated from the slopes of the temperature curves shown in Figures 2 and 3. For a given set of process parameters, since the temperature fluctuated rapidly as the material experienced multiple melting and solidification cycles, the cooling rates encountered for each solidification cycle were different, as can be observed from Figure 5. We are more concerned with the cooling rates at the solidification temperature and below than at the peak temperature in the weld pool. It already has been shown that for the same welding conditions, the experienced thermal cycle and, consequently, the cooling rate may differ at different locations. In addition, the calculations indicate that at any given location, the weld metal experienced successive melting and solidification

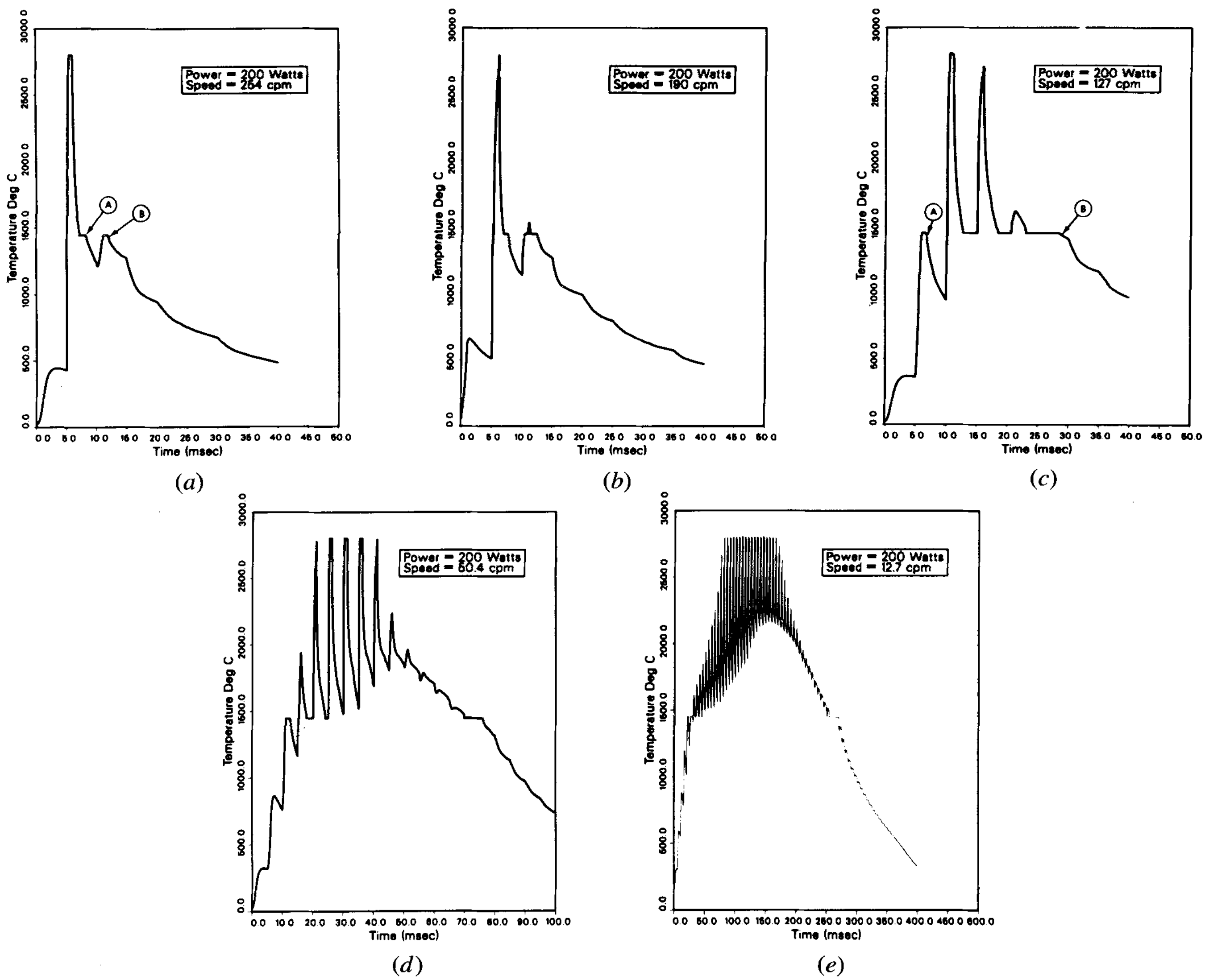

Fig. 2—Calculated weld thermal cycles at the surface for a power level of 200 W: (*a*) 254 cm/min; (*b*) 190 cm/min; (*c*) 127 cm/min; (*d*) 50.4 cm/min; and (*e*) 12.7 cm/min. Marked points are referenced when describing cooling rates.

at cooling rates that may vary with each solidification cycle. For example, for a power level of 200 watts and welding speed of 254 cm/minute, the calculated cooling rates at 1450 °C (melting point) during the two solidification cycles (indicated by the arrows A and B in Figure 2(a)) were 3.2×10^5 °C/second and 3.0×10^5 °C/second. Corresponding to Figure 2(c) (200 watts and 127 cm/minute), the cooling rates at 1450 °C for the two solidification cycles (indicated by the arrows A and B) were 4.3×10^5 °C/second and 0.56×10^5 °C/second. In short, for a given power level and welding speed, the weld metal experienced a range of cooling rates that is different for each location and thermal cycle. Therefore, only general trends in the variation of cooling rates will be reported here.

In general, for the conditions investigated, the cooling rates at the solidification temperature increased with increasing scanning speed. For the two power levels considered, the cooling rates at the solidification temperature were higher for the lower power level at all welding speeds. The calculated cooling rates ranged from 0.56 to 3.2×10^5 °C/second for the higher power level (200 W). At the lower power level (120 W), the calculated cooling rates ranged from 1.0 to 8.8×10^5 °C/second.

The results also indicate that, depending on the process parameters and location of the monitoring point, the post-solidification cooling rates experienced by the weld metal can increase with decrease in temperature making it very difficult to obtain general trends in the variation of cooling rates with temperature. For example, during the last solidification cycle in Figure 3(d) (120 watts, 50.4 cm/minute), the weld metal experienced a cooling rate of 0.17×10^5 °C/second at 1200 °C (indicated by the arrow A) which subsequently increased to 0.55×10^5 °C/second at 1100 °C (indicated by the arrow B) and then decreased slightly to 0.52×10^5 °C/second at 1000 °C (indicated by the arrow C). Similarly, in Figure 3(c) (120 watts, 127 cm/minute), the calculated cooling rates at the same three temperatures were 0.71, 1.8, and 0.62×10^5 °C/second (indicated by the arrows A, B, and C), respectively.

Typically, the temperature gradient, which is the driving force for heat flow, and the capacity of the adjacent

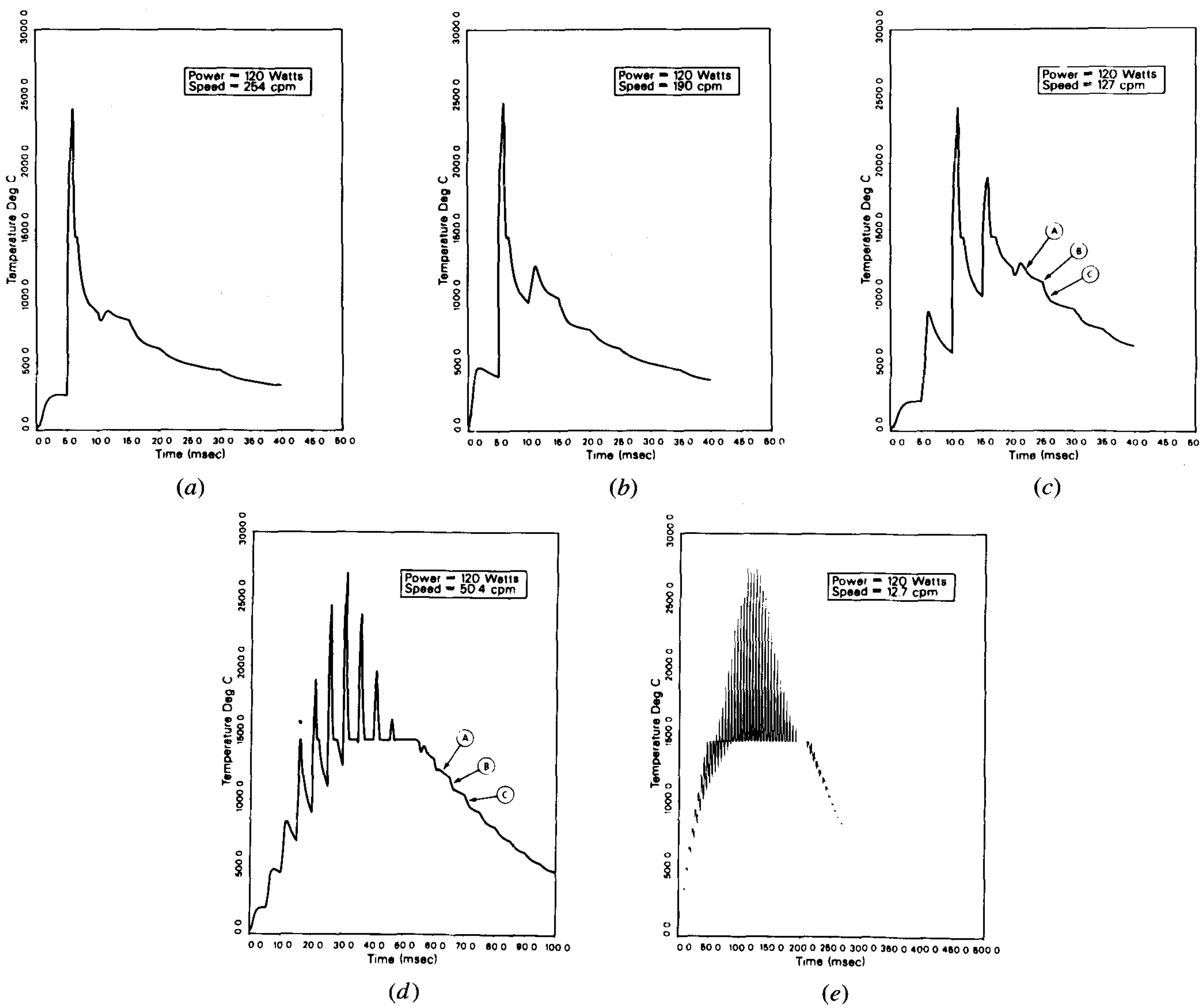

Fig. 3—Calculated weld thermal cycles at the surface for a power level of 120 W: (*a*) 254 cm/min; (*b*) 190 cm/min; (*c*) 127 cm/min; (*d*) 50.4 cm/min; and (*e*) 12.7 cm/min. Marked points are referenced when describing cooling rates.

base metal to remove heat from the fusion zone determine how fast a material cools through a particular temperature regime. Generally, for a continuous, moving heat source, the cooling rates are an exponentially decaying function of time. However, the intermittent heat source during pulsed laser welding produced several temperature peaks as the material was successively irradiated with the laser beam. During the off time of the laser beam, the material cooled rapidly until the next irradiation, at which time a particular location may experience heating (Figures 2 and 3), if the location is close enough to the laser beam. The additional heat supplied to the material altered the prevailing temperature gradient by changing the temperature of the location as well as the adjacent base metal, thereby altering the cooling rates.

Without direct measurements of the cooling rates during laser welding, the calculated results can be verified qualitatively by using a previously established relationship between dendrite arm spacings and cooling rates. Brower *et al.*[25] experimentally investigated the effect of cooling rates on the structure of splat quenched 440C, Fe-25 pct Ni, and Fe-0.027 pct Si, 0.019 pct O alloys. Their results showed that the secondary arm spacing varied linearly with cooling rate to an exponent that lies between -0.33 and -0.5. Previous work by the authors has shown that the solidification structures of the splat quenched specimens are similar to those observed during laser welding.[10] A limited number of measurements of secondary arm spacings for the solidification of AISI 201 made by Abdulgader also are available in the literature.[17] The results in the literature[17,25] indicate that the relation between the dendritic arm spacings and cooling rate is given by

$$d = 60\,\dot{T}^{-0.41} \qquad [4]$$

where d is the secondary dendrite arm spacing and $\dot{T}$ is the cooling rate. This relationship was used to derive the secondary dendrite arm spacings from the calculated cooling rates. For the range of calculated cooling rates predicted for the present welding conditions, the calculated dendrite arm spacings ranged from 0.22 to 0.63 μm.

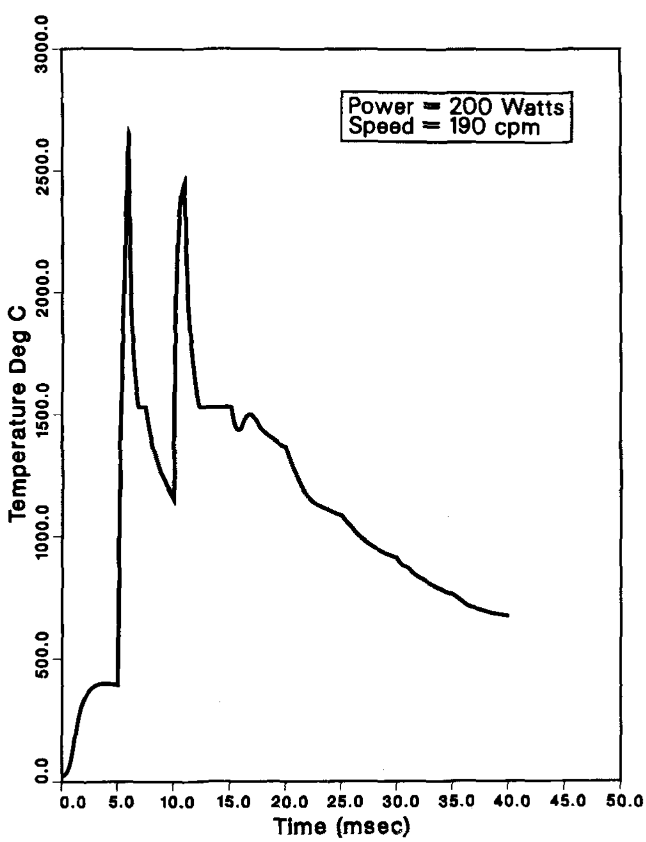

Fig. 4—Calculated weld thermal cycles at the surface for a power level of 200 W and welding speed of 190 cm/min. The conditions are the same as in Fig. 2(b), but the location of the monitoring point with respect to the beam position is different.

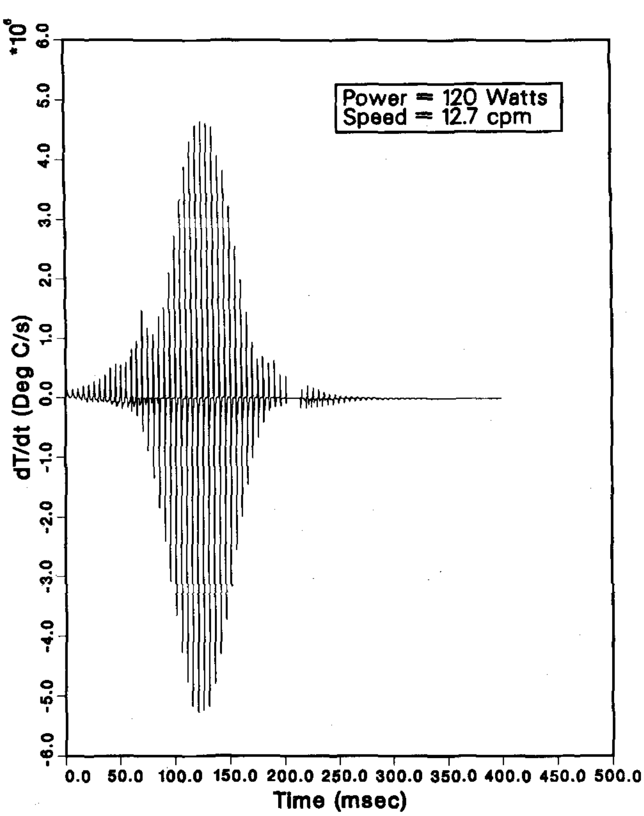

Fig. 5—Representative plot of calculated rate of change of temperature as a function of time (corresponds to Figure 3(e)).

For most welding conditions investigated, microstructural analysis showed no discernible secondary dendrite arms in the solidification structure, preventing a comparison with the model predictions. However, in Type 310 and 316 alloys, for some of the welding conditions, the secondary dendrite arm spacings were discernible and could be measured using optical microscopy. Reasonable care was taken to ensure that, whenever possible, the secondary arm spacings were measured at the same locations with respect to the surface. The measured secondary dendrite arm spacings varied from 0.21 to 0.73 μm and are plotted as a function of the calculated cooling rates in Figure 6. The results indicate good agreement between the measured and the expected secondary dendrite arm spacings.[17,25] Furthermore, the predictions are in good agreement with other reported estimates of cooling rates during laser welding.[26]

Comparing the calculated cooling rates to the estimated cooling rates reported earlier,[12] the present calculations, though lower, are within the same order of magnitude. For example, for the various welding conditions investigated, the calculated cooling rates ranged from 0.56 to 8.8 $\times$ 10^5 °C/second in comparison to the previous estimate of 0.7 $\times$ 10^5 °C/second to 2.3 $\times$ 10^6 °C/second. Even though the range of cooling rates predicted is within the same order of magnitude, the present analysis shows the large expected influence of pulsing and expected nonuniformity in cooling rates as the weld metal experiences multiple solidification cycles. Since the cooling rate during solidification can significantly influence the solidification mode and structure,[12] one can expect different solidification modes and nonuniform structures. In the next sections, the experimentally observed solidification modes and structures are explained in light of the calculated thermal cycles and cooling rates.

C. *Macrostructure*

It already has been discussed that welding using a pulsed laser gives rise to unique thermal cycles and cooling rates during solidification. The effect of rapid temperature excursions due to pulsing on the solidification structure can be understood by considering the schematic section of a bead on plate weld given in Figure 7. Figure 7(a) is the

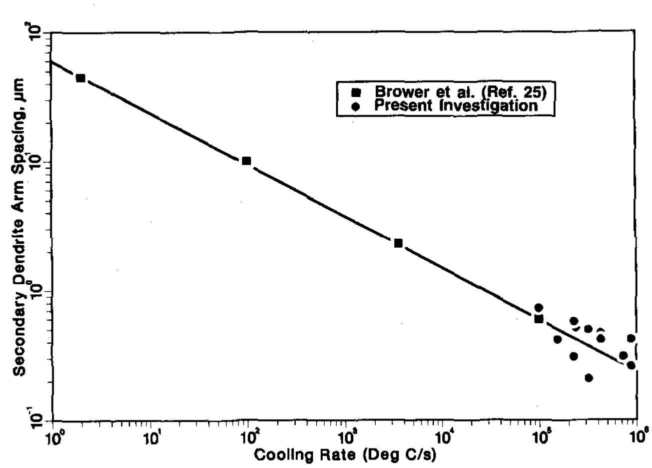

Fig. 6—Comparison of calculated and experimentally observed secondary dendrite arm spacings as a function of calculated cooling rates.

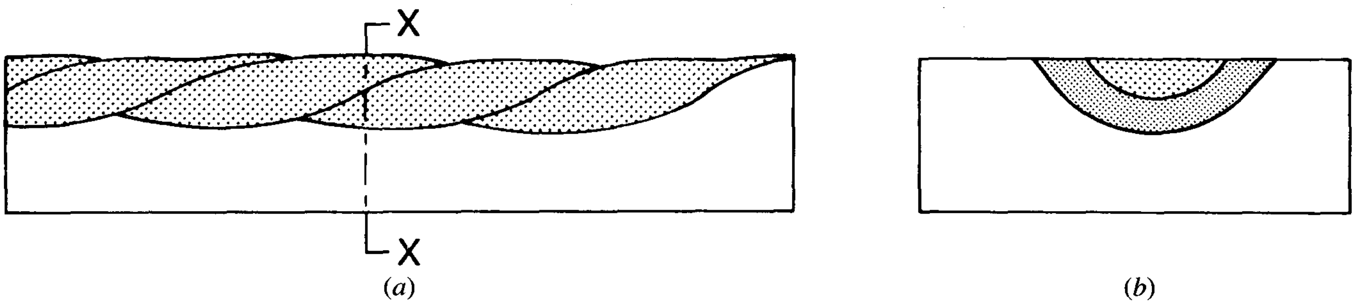

Fig. 7—Schematic section of a bead on plate weld for pulsed laser welding: (a) longitudinal section; (b) transverse section.

longitudinal section of the weld metal, indicating the overlapping puddles due to successive laser pulses. Figure 7(b) shows the transverse section along a plane X-X, indicating multiple melt fronts, the exact number depending on the power, the scanning speeds, and the location of the transverse section. For example, at a speed of 190 cm/minute and power level of 120 W, as shown in Figure 8, bands are found in both the longitudinal and transverse sections of the weld metal, corresponding to the different solidification fronts, where individual puddles had overlapped. Since the cooling rate may vary at the different solidification fronts, vastly different solidification modes and structures can occur. In addition, the weld metal can undergo extensive microstructural modification (solid-state transformation), as the previously solidified fusion zone becomes the heat affected zone of the next molten puddle. Therefore, the observed solidification structure is the cumulative result of multiple, complex laser-metal interactions.

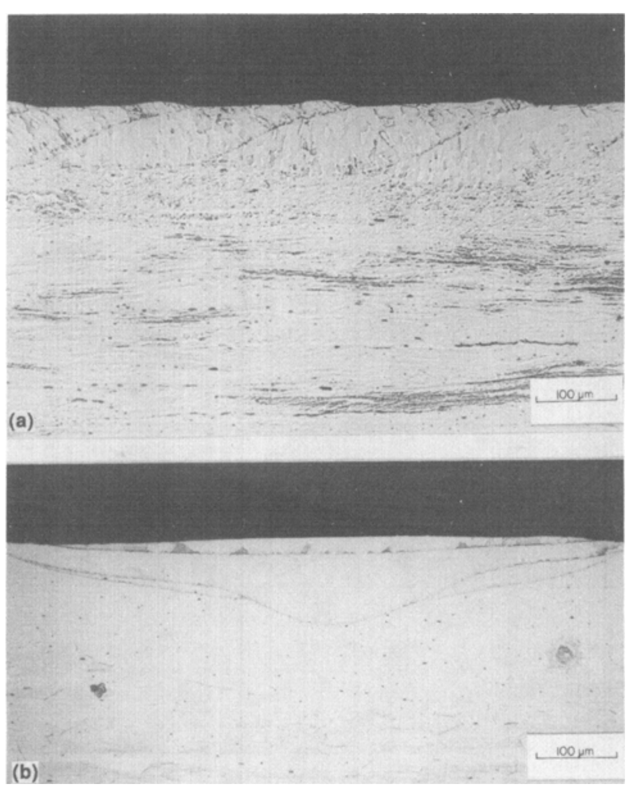

Fig. 8—Macrographs of type 308 laser beam welds (120 W, 190 cm/min), indicating the solidification fronts for successive weld puddles: (a) longitudinal section; (b) transverse section.

Consider Figure 2(a), which shows the calculated thermal cycle for a power level of 200 watts and scanning speed of 254 cm/minute. The calculations in Figure 2(a) indicate that a point on the surface of the metal experienced two successive melting and solidification cycles as the laser scanned along the direction of the weld. A representative transverse macrograph indicating the melt fronts corresponding to the thermal cycle is shown in Figure 9. Two predominant melt fronts where successive puddles overlapped are shown, as expected. In addition, the macrograph indicates the remnants of another melt front to the sides of the fusion zone. It must be pointed out that the experimentally observed melt fronts are expected to be very sensitive to the exact location of the fusion zone from which the metallographic section was prepared. At a location different from the monitoring point, the material can experience additional melting and solidification cycles, as shown in Figure 9.

The calculated thermal cycle corresponding to a power level of 200 watts and scanning speed of 50.4 cm/minute is shown in Figure 2(d). Repeated melting and solidification cycles are predicted, but since the scanning speed is low, successive laser metal interactions take place within a very small area, causing the weld metal to remain molten even during the off time of the beam pulsing and resulting in the production of virtually continuous welds. This is in agreement with the experimental observation of these welds (Figure 10). With decreasing scanning speed (12.7 cm/minute), this effect was found to be more pronounced, and it resulted in an increased depth of penetration. For the laser welding conditions investigated, the depth of penetration ranged from 0.1 mm at low power and fast speed to full penetration (0.51 mm) for slow speeds.

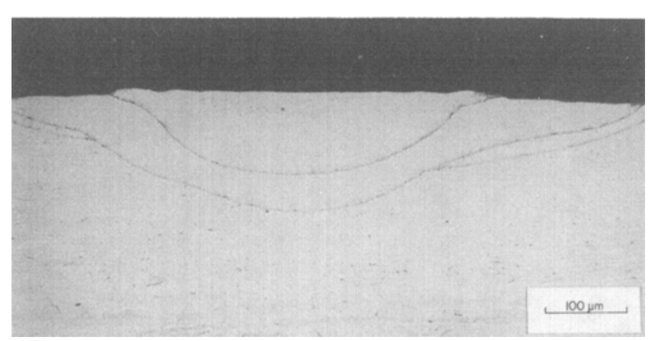

Fig. 9—Transverse macrograph of the fusion zone, indicating the interfaces where successive puddles overlapped for power level of 200 W and welding speed of 254 cm/min.

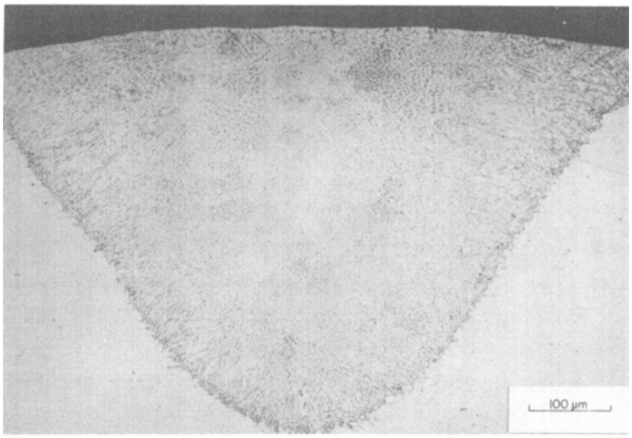

Fig. 10 — Representative micrograph of the fusion zone, indicating a virtually continuous weld bead: 200 W, 50.4 cm/min.

D. *Microstructure*

For many of the compositions investigated, extensive microstructural modifications were observed during laser welding. Observed laser weld microstructures were categorized by optical microscopy into one of the three groups: fully austenitic (γ); fully ferritic (δ); and duplex austenite + ferrite ($\gamma + \delta$). Following conventional welding, the alloy types 304, 308, 312, and 316 contain duplex ($\gamma + \delta$) microstructure and solidify in the primary ferritic mode. For both of the laser powers investigated, at slow laser welding speeds (correspondingly low cooling rates), the same duplex ($\gamma + \delta$) structures were observed. However, for alloys 304, 308, and 316, as the welding speed increased, a decrease in ferrite content was observed, ultimately resulting in fully austenitic microstructures at the highest welding speeds.[12] An example of these results is given in Figure 11. The observed fully austenitic microstructure at the high welding speeds or associated high cooling rates may be attributed to a change in mode of solidification from primary ferritic to one of primary austenitic.[10] The interdendritic regions in Figure 11(f) appear dark due to the etching effect. No second phase, and in particular no ferrite, was observed in the microstructure. Alloy 312 showed a different type of structural dependence on the cooling rate. This alloy, following conventional welding, may typically contain 40 pct ferrite after solidifying in the primary ferrite mode. However, during laser welding, as the welding speed increased, the ferrite content increased until a fully ferritic microstructure was obtained at the higher welding speeds.[12]

The type 310 steel contains a fully austenitic microstructure after welding by conventional means. Also, it has been found to solidify in a primary austenitic mode.[7] During laser welding, this alloy showed no evidence of microstructural modification. A fully austenitic microstructure was found for all conditions examined.

The thermal conditions experienced by the weldment may have a significant effect on the solid-state transformations that occur after solidification. For example, at slow speeds (127 cm/minute) and high power level (200 W), as shown in Figure 12, duplex austenite plus ferrite areas are found in the immediate vicinity of the overlapping weld pool front in samples that otherwise are fully austenitic. This can be understood, if one realizes that these regions near the solidification front are

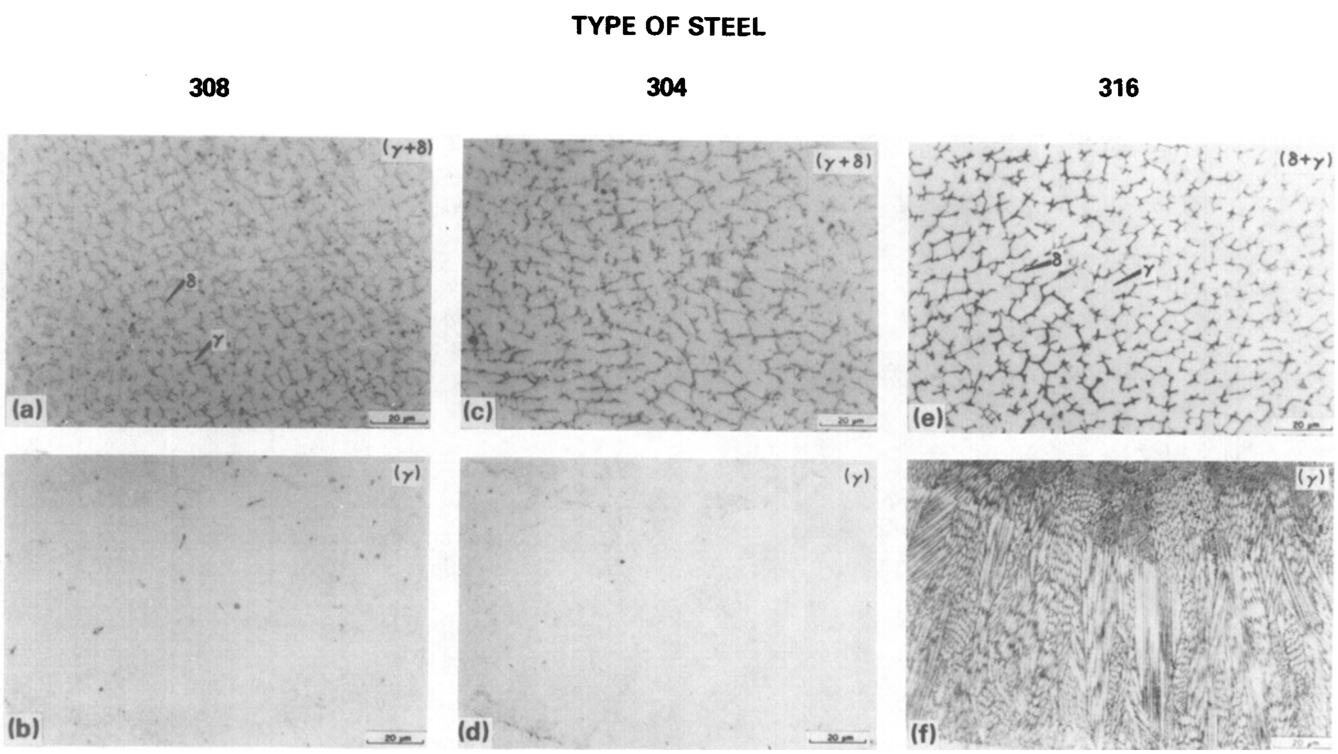

Fig. 11 — Microstructure of types 308, 304, and 316 laser beam welds with increasing welding speed: (*a*), (*c*), (*e*) 12.7 cm/min; (*b*), (*d*), (*f*) 190 cm/min.

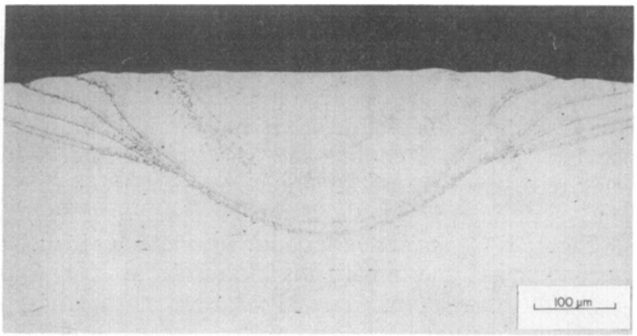

Fig. 12 — Microstructure of type 308 laser beam welds (200 W, 50.4 cm/min), showing solid state transformation resulting from the pulsed laser beam.

reheated during subsequent pulses to nearly the solidus temperature, and, therefore, may undergo a solid-state transformation of austenite to ferrite. At such elevated temperatures, the ferrite is the stable phase. If ferrite does form during this elevated temperature exposure, it is likely to remain intact during cooling since the cooling rates are expected to be very high (10^5 °C/second), thereby avoiding the reverse ferrite to austenite transformation. For slower speeds, since welds similar to those expected of continuous wave laser are obtained (multiple melt fronts are absent) as shown in Figure 10, the structure is more uniform and showed no evidence of solid-state transformation.

In addition, the thermal cycling also may have a noticeable effect on other solid-state transformations. Some of the alloys welded at high speeds (alloy 316; Reference 12) showed a very fine microstructure indicative of recrystallization. Such recrystallized microstructures can be explained readily by the temperature excursions predicted in the present thermal analysis. With the development of residual stresses during cooling, and subsequent exposure to high temperatures as shown in Figure 3(b), recrystallization certainly is possible.

Finally, as shown in Figure 13, high dislocation densities were found in some of the materials, particularly after welding at relatively high speeds. Once again, thermal cycling as predicted by the calculations can be responsible for the development of large residual stresses, and, therefore, high dislocation densities.

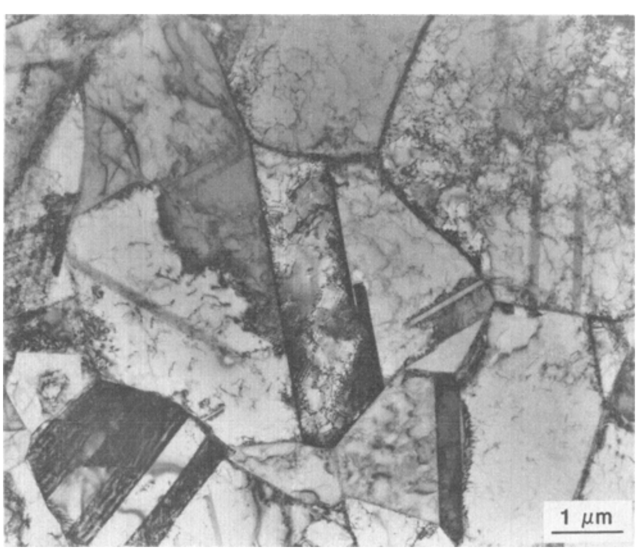

Fig. 13 — TEM micrograph showing fully austenitic microstructure of type 316 stainless steel, laser beam welded at 50.4 cm/min, and containing a high dislocation density.

V. CONCLUSIONS

A detailed experimental and theoretical study of the effect of pulsed laser welding on the thermal response of austenitic stainless steels was carried out. The results and conclusions are summarized as follows:

1. A detailed analysis has been made of the heat transfer that occurs during pulsed laser welding. The model described offers a useful means of obtaining the complex temperature distribution and cooling rates that occur during pulsed laser welding.
2. The predictions indicate that the weld metal experiences multiple melting and solidification that are sensitive to the power level and scanning speeds. At low speeds, the results indicate that the weld metal remains molten even during the off time of the laser beam. The peak temperatures observed in the molten puddle were at or very close to the boiling point for all the conditions investigated.
3. The predicted cooling rates ranged from 0.56 to 8.8 $\times$ 10^5 °C/second. At the solidification temperature, the cooling rates increased with increasing scanning speed and decreasing power.
4. For the welding conditions studied, the theoretically predicted secondary dendrite arm spacings ranged from 0.22 to 0.63 μm. These results are in agreement with the experimentally observed secondary dendrite arm spacings.
5. The solidification of the weld metal at these high cooling rates produced significantly different microstructures than those found after conventional welding. The observed microstructures were sensitive to the cooling rate and ranged from duplex $\gamma + \delta$ to fully austenitic or fully ferritic.
6. There was microstructural evidence of significant solid-state transformation in the fusion zone. The thermal cycles that are predicted in the present analysis can explain the observed duplex austenite plus ferrite structure in the immediate vicinity of the overlapping puddles, apparent recrystallized microstructures, and high dislocation densities in the fusion zone.

APPENDIX

Definition of Symbols

c_p	specific heat
$\mathbf{g}$	gravitational constant
H	enthalpy
p	pressure
$\mathbf{q}_s$	heat transfer flux
Q	laser power
r_b	effective radius of heat flux
t	time
$\dot{T}$	cooling rate
u	x-component of velocity
v	y-component of velocity

y_d	absorption depth
δ_k	Kroenecker delta
ε	absorptivity
γ	surface tension coefficient
κ	thermal conductivity
ν	kinematic viscosity
ρ	density

Method of Solution

The governing equations and the boundary conditions were expressed in finite difference form and solved on a line-by-line basis utilizing the tridiagonal matrix algorithm (TDMA). First, the energy equation was solved for the entire computational domain using the TDMA, starting with the known values of enthalpy corresponding to room temperature. The pool geometry was determined based on the calculated values of enthalpies and the momentum equations solved within this molten pool. The continuity equation was satisfied identically to within 1 pct of a specified reference mass divergence. The calculated enthalpies and velocities were updated for each time level. The position of the beam was adjusted with every time step.

The mathematical model used a 50×40 nonuniform, fixed rectangular grid system for the calculation of enthalpies and velocities. Finer grids were utilized near the monitoring location, and relatively coarse grids were used far away from the monitoring location. Similarly, finer grids were utilized near the surface of the specimen where steep velocity gradients are induced by the interfacial tension gradient. Far away from the surface, a relatively coarse grid was utilized.

Stability, Convergence, and Accuracy of the Numerical Scheme

Unconditional stability was obtained since the solution procedure was an implicit one. Thus, an unduly small time increment mandated by an explicit scheme is not required.

The maximum permissible mass imbalance in each cell was 1 pct of a reference mass divergence. The value of this reference divergence was taken as 5000 seconds^{-1} for a minimum grid spacing of 0.001 cm and a typical velocity difference of 5 cm/second. The momentum and enthalpy convergence was achieved when u, v, and H satisfied the momentum, continuity, and enthalpy equations simultaneously. The maximum permissible relative difference between the velocities at each location obtained from the solution of the continuity and momentum equations was 0.01.

A parametric study indicated that time increments less than 0.0005 seconds did not give any different results for a minimum grid spacing of 0.001 cm. Hence, for all calculations, the time increment was taken as 0.0005 seconds.

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